Network Devices Overview

What it is

Network devices are purpose-built hardware (or virtualized equivalents) operating at distinct layers of the OSI model, each performing a specific forwarding, segmentation, or translation function. The five foundational device types — **hub**, **bridge**, **switch**, **router**, and **gateway** — differ in which layer they operate at, what address information they inspect, and how they forward traffic.

Why it matters

Every networked application a developer writes ultimately crosses multiple device types. Understanding where packets are broadcast, filtered, or routed determines why latency exists, why broadcasts storm, why VLANs work, and why a "router" in your home is actually several devices in one box.

How it actually works

Layer 1 — Hub

A **hub** is a physical-layer repeater. It has no intelligence beyond signal regeneration. When a frame arrives on any port, the hub repeats the electrical signal out every other port simultaneously. There is no MAC address table, no collision domain segmentation per port — all ports share **one collision domain and one broadcast domain**.

- Operates at **OSI Layer 1** (Physical).
- Repeats incoming bits to all other ports.
- Cannot filter, cannot learn MACs, cannot isolate traffic.
- Largely obsolete; replaced by switches even at the cheapest price points.

Layer 2 — Bridge

A **bridge** connects two (or more) network segments and selectively forwards frames between them based on **MAC addresses**. It learns which MACs live on which side by inspecting source addresses of incoming frames, populating a MAC address table (also called a CAM table).

- Operates at **OSI Layer 2** (Data Link).
- Reduces collision domain size by separating segments.
- Forwarding decision uses destination MAC; unknown unicast is flooded.
- The conceptual ancestor of the modern switch. A bridge with multiple ports *is* effectively a multiport bridge, which is exactly what a switch is.

Layer 2 — Switch

A **switch** is a multiport bridge. It maintains a **MAC address table** mapping MAC addresses to physical switch ports, populated dynamically through a learning process:

1. On frame arrival, the switch reads the **source MAC** and records it against the ingress port (with a timestamp/aging timer, typically **300 seconds** on Cisco defaults).

2. It looks up the **destination MAC** in the table.
3. Three outcomes:
    - **Known unicast**: frame forwarded only out the learned port.
    - **Unknown unicast**: frame flooded out all ports except the ingress port.
    - **Broadcast/multicast**: flooded out all ports except the ingress port (unless IGMP snooping or similar features are active).

- Operates at **OSI Layer 2** by default; **Layer 3 switches** add routing functionality (see below).
- Each switch port is its own **collision domain**; the entire switch is a single **broadcast domain** unless VLANs are configured.
- Forwarding is typically done in hardware via an **ASIC** (Application-Specific Integrated Circuit) for wire-speed performance.
- Common forwarding modes: **store-and-forward** (entire frame buffered, CRC checked before forwarding — adds latency but catches errors) and **cut-through** (begins forwarding as soon as destination MAC is read — lower latency but forwards corrupted frames). Modern switches often use **fragment-free** (a variant that waits for the first 64 bytes, the minimum Ethernet frame size, to catch most collisions) as a compromise.

Layer 3 — Router

A **router** connects **different IP networks/subnets** and forwards packets based on **destination IP address** using a routing table.

- Operates at **OSI Layer 3** (Network).
- Maintains a **routing table** populated via static routes or dynamic routing protocols (OSPF, BGP, RIP, EIGRP).
- Each router interface is a separate **broadcast domain** and a separate **collision domain**.
- Performs **TTL decrement** on every forwarded packet; drops packets when TTL reaches 0 and sends an **ICMP Time Exceeded (type 11)** to the source.
- Performs **IP header checksum recalculation** since TTL changed.
- Performs **MTU handling**: if outbound interface MTU < packet size and the Don't Fragment (DF) bit is set, the router sends **ICMP Destination Unreachable (type 3, code 4) — "fragmentation needed and DF set"** back to the source. This is the basis of **Path MTU Discovery (PMTUD)**, defined in **RFC 1191**.
- Routers segment broadcast domains — broadcasts do not cross a routed boundary.

A **Layer 3 switch** is essentially a router implemented in switch hardware (ASIC-based forwarding) with high port density. It performs the same logical function as a router but is optimized for intra-datacenter or campus traffic.

Layer 3+ — Gateway

A **gateway** is a device or software that translates between **different network protocols or architectures**. The term is overloaded:

1. **Default gateway**: the router IP address a host uses to send traffic destined for off-subnet destinations. The host's IP stack checks if the destination is in its own subnet; if not, it ARPs for the default gateway's MAC and sends the frame to it.
2. **Application gateway / protocol gateway**: translates between protocols — e.g., email gateway between X.400 and SMTP, VoIP gateway between SIP and PSTN, or an API gateway that translates

between HTTP/REST and a backend protocol.

- Operates at any layer depending on function; can span OSI layers 4–7 when translating protocols.
- The default gateway is the most common meaning for developers.

Modem

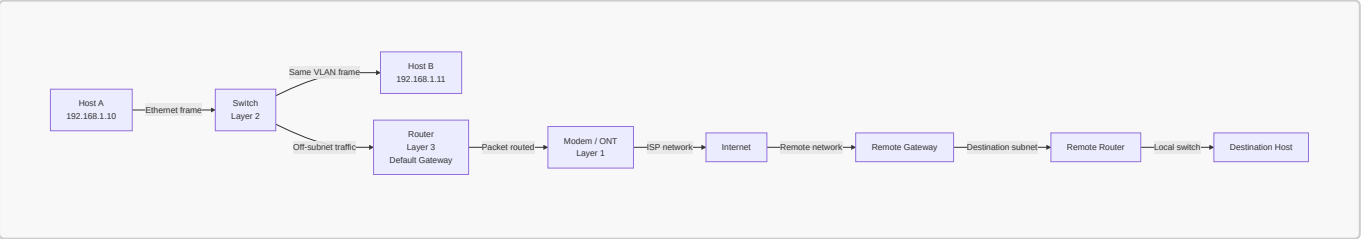
A **modem** (modulator-demodulator) converts digital data into a signal format suitable for the physical transmission medium — and vice versa. The modem is a **Layer 1 device** that adapts the digital interface (typically Ethernet) to the analog or carrier signal of the WAN medium.

- **DSL modem**: uses discrete multitone modulation over telephone lines; frequencies above ~4 kHz carry data.
- **Cable modem**: uses **DOCSIS** (Data Over Cable Service Interface Specification) over coaxial cable TV infrastructure, sharing bandwidth with other users in the segment (HFC — hybrid fiber-coax).
- **Fiber ONT/ONU**: technically a media converter rather than a "modem" in the traditional sense; converts optical signals to Ethernet.
- Modern home "modems" are often **gateway combos** that bundle a modem + router + switch + Wi-Fi AP + firewall in one physical device — which is why so many developers conflate these device types.

Address types per device

Device	Examines	Forwards based on	Learns
Hub	Nothing	N/A — repeats all	Nothing
Bridge	Destination MAC	MAC table	Source MAC → port
Switch	Destination MAC	MAC table	Source MAC → port
Router	Destination IP	Routing table	Routes via protocols
Gateway	Varies by purpose	Protocol translation rules	Varies

Architecture / flow



Key terms

- **Collision domain** — A network segment where simultaneous transmissions collide; switches and routers isolate these per port/interface.
- **Broadcast domain** — The set of devices that receive a broadcast frame; routers break broadcast domains, switches do not (without VLANs).
- **CAM table** — Content-Addressable Memory table inside a switch; the MAC address → port mapping used for L2 forwarding.

- **MTU** — Maximum Transmission Unit; largest IP packet size (default **1500 bytes** for Ethernet) that can pass without fragmentation.
- **ICMP** — Internet Control Message Protocol; carries diagnostic messages like "destination unreachable" and "time exceeded."

Example

A typical home gateway's logical topology, as you'd see it in the device's admin UI:

```
WAN interface (modem) ──► Public IP assigned via DHCP/PPPoe
        │
        ▼
   Router (Layer 3) ── NAT, DHCP server, firewall
        │
        ▼
   Switch (Layer 2) ── Ports 1–4, internal VLAN 1 (192.168.1.0/24)
        │
        ├──► Wi-Fi AP (often bridged into same VLAN)
        │
        └──► Ethernet ports
```

This single box performs modem, router, switch, AP, and firewall functions — which is precisely why "router" in a consumer context usually means several devices running together.

Common mistakes

- Calling a home gateway "the router" and assuming the Wi-Fi access point, switch, and modem are separate — they're not, but their failure modes are.
- Assuming switches eliminate broadcasts — they don't, only routers (or VLAN-interfaces / Layer 3 switches) do.
- Forgetting that the host itself makes the routing decision first: if the destination IP is on a different subnet, the host sends the frame to its **default gateway's MAC**, not to the destination host's MAC. The destination IP in the packet stays unchanged through the routed path; only Layer 2 addresses are rewritten hop-by-hop.
- Assuming all switches do Layer 3 — a basic "unmanaged switch" is pure Layer 2 and will drop routed traffic or ignore IP headers entirely.

Lesser-known internals

- **Store-and-forward vs cut-through** is not just a latency choice — store-and-forward catches CRC errors and runts before they consume bandwidth, while cut-through can propagate a bad frame across many switches before the error is detected.
- **Switches and broadcast storms**: without **STP (Spanning Tree Protocol, IEEE 802.1D)** or its faster successor **RSTP (802.1w)**, redundant Layer 2 paths create forwarding loops where broadcasts multiply exponentially and saturate the network in seconds. This is why STP is enabled by default on managed switches.
- **Asymmetric routing** breaks stateful firewalls and some NAT setups: if traffic from host A to server S goes out one ISP link and the return traffic comes in another, the firewall may drop the reply because

it never saw the original outbound packet on that interface.

Topics to explore further

- **VLANs and 802.1Q tagging** — how switches partition broadcast domains with a 4-byte tag inserted into the Ethernet frame.
- **Spanning Tree Protocol variants** — MSTP (802.1s), PVST, and how loops are prevented.
- **NAT and PAT** — how a single public IP maps to many private IPs, and why it breaks protocols that embed addresses in payloads.
- **BGP and OSPF** — the dynamic routing protocols that populate routing tables at ISP and enterprise scale.

Learn next

- **OSI and TCP/IP models** — the layered framework that defines what each device actually does. *Direct foundation for this topic.*
- **Ethernet framing and MAC addressing** — the exact byte layout switches parse. *Explains how a switch's CAM table works.*
- **IP addressing and subnetting** — required to understand router forwarding decisions. *Defines the router's forwarding logic.*